



Vishal S R 250701844-2025-CSE ▾

V2**Started on** Monday, 24 August 2026, 10:36 PM**State** Finished**Completed on** Monday, 24 August 2026, 11:01 PM**Time taken** 24 mins 55 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int binarySearch(int arr[], int low, int high, int target)
4  {
5      if (low > high)
6          return -1;
7
8      int mid = (low + high) / 2;
9
10     if (arr[mid] == target)
11         return mid;
12
13     if (arr[mid] < target)
14         return binarySearch(arr, mid + 1, high, target);
15
16     return binarySearch(arr, low, mid - 1, target);
17 }
18
19 int findPair(int arr[], int n, int x, int index)
20 {
21
22     if (index >= n)
23         return 0;
24
25     int required = x - arr[index];
26
27     int pos = binarySearch(arr, 0, n - 1, required);
28
29     if (pos != -1 && pos != index)
30     {
31         printf("%d\n", arr[index]);
32         printf("%d\n", arr[pos]);
33         return 1;
34     }
35
36
37     return findPair(arr, n, x, index + 1);
38 }
39
40 int main()
41 {
42     int n, x;
43
44     scanf("%d", &n);
45
46     int arr[n];
47
48     for (int i = 0; i < n; i++)
49     {
50         scanf("%d", &arr[i]);
51     }
52

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)